

Bio-inspired Spherical Joints for a Soft Articulated Artificial Hand

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Abstract—The introduction of soft robotics technologies represents a promising direction to enhance system robustness and natural interaction. However, the inherent flexibility in joint designs can pose challenges in achieving both precise and repeatable gripping. This work introduces a soft artificial hand design with constrained ball joints, inspired by the anatomical structure of bones and tendons in the human hand. The key innovation lies in the integration of selectable stiffness levels in the spherical joints, enabling distinct stiffness characteristics to achieve both power and precision grasps. Through a set of contact simulations, this study characterizes different behaviors in the finger model according to the stiffness level of its joints. The final hand design encompasses 15 degrees of freedom (DoF) actuated by a unique tendon and a single motor, and a passive proximal thumb joint. The developed prototype was preliminarily tested to assess its manipulation taxonomy and demonstrated the ability to grasp everyday objects in different configurations. Results show the potential of the proposed joint design in achieving a trade-off between precision and adaptability in grasping.

I. INTRODUCTION

The human hand represents one of the most intricate mechanisms of the human body for its structural complexity and capabilities [1]. Since the sixteenth century, scientific and engineering efforts have been dedicated to replicating the richness and complexity of the human hand in artificial devices [2], [3], [4]. Traditional design methodologies have focused on meticulously imitating both the appearance and dexterity of human hands. These endeavors often involve sophisticated designs that integrate numerous actuators and sensors [5], [6]. In these systems, finger design commonly adopts rigid joints, which consist of fixed mechanical elements (e.g., pins) to connect the links. Furthermore, the preferred transmission architecture is typically fully actuated, allowing for direct control of each joint through a dedicated motor. While this approach ensures precision and repeatable grasps, it comes with drawbacks, including increased weight, unnatural interactions, and lack of robustness [7].

In recent years, advancements in the development of artificial hands have embraced numerous innovative approaches inspired by biological structures [8], [9], that are both effective and reliable. Recent studies explore diverse joint solutions that lead to the design of fingers with diverse properties, catering to various functionalities [10], [11], [12]. These

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Fig. 1. Primary grasp types in free closure, including power, precision and lateral grasps. Each row illustrates a distinct grasp, with the first and second columns depicting the hand in its initial and final grasp geometry, respectively. Note that the thumb can be passively moved to achieve different contact scenarios, defining each grasp type. The third column showcases these grasps with an object.

novel artificial hands transition from rigid and machine-like designs to solutions incorporating softness into both the mechanics of articulations and actuation systems. Over the past century, numerous variations have been extensively explored in the literature (a review in [3]), characterized by a substantial reduction in the number of actuators and sensors. For instance, various systems have explored simplified actuation architectures, yielding promising outcomes in terms of usability and practicality [13], [14], [15]. Alternative strategies draw inspiration from neuroscientific studies, proposing that the human sensory-motor system effectively manages complexity through a stratified set of motor primitives, also referred to as postural synergies [16]. Roboticians are capable to design and integrate prescriptive synergies based on the kinematic data extracted from human experiments. Therefore, prescriptive postural synergies are the predetermined trajectories implemented into the controller to produce an action with the robot, representing an intrinsic characteristic of the system's actuation architecture. While retaining the advantages of anthropomorphic designs, a principled simplification in the actuation architecture can significantly reduce system complexity, encompassing the number of motors, sensors, lines of code required for programming, and thereby, control complexity. In our prior research, we investigated the translation of the soft-synergy model [17] into a physical system [18], resulting in enhanced movement naturalness and

the intuitiveness of artificial limbs [19], [20], [21].

Although the adoption of soft robotic technologies exhibit exceptional performance in power grasp scenarios [14], limitations often encounter when precision gripping is required, primarily due to the inherent flexibility of their joint design [22]. To overcome this constraint in critical grasping actions, this research introduces the design of an artificial hand based on soft articulated robotics technologies, featuring spherical joints with selectable stiffness levels, termed as constrained artificial ball joints. This design approach draws inspiration from the anatomical structure of bones and tendons in the human hand. The proposed system, encompassing 15 degrees of freedom (DoF) and a single actuator, primarily operates along the first postural synergy mode using a single tendon. The incorporation of constrained ball joints in our design allows the selection of distinct stiffness characteristics at each finger or phalangeal joint. This novel feature facilitates a more repeatable finger trajectory in stiffer joints, enabling precision grasps, while simultaneously permitting lateral displacement in softer joints. Furthermore, this design allows passive adjustments to the configuration of the metacarpal thumb joint, enabling various grasping configurations, such as power, precision, and lateral grasp (Fig. 1). This study first demonstrates different stiffness behaviors in the finger design through various contact simulations and preliminary testing with the physical system. To quantitatively assess human manipulation, taxonomies are frequently employed to formalize the spectrum of potential grasps, including robotic end-effectors [23]. We used a frequently referenced framework [24] that categorizes all human grasps into either power or precision categories.

II. DESIGN

A. Human hand anatomy and bio-inspiration

Drawing inspiration from human hand anatomy [1], we designed a novel soft articulated artificial hand featuring four bio-inspired structures. Dense stiff components replicate bones, providing crucial structural support. Their movement is allowed by muscles and tendons, translated into our system through a tendon-based actuation architecture powered by a motor. In our design, elastic elements are used to replace ligaments, the fibrous tissues that enclose joints. Each robotic finger structure presents three spherical joints, replicating the distal (DIP), proximal (PIP), and metacarpal (MCP) joints (Fig. 2), commonly known as ball-and-socket joints. Drawing inspiration from synovial joints, our design facilitates fluid and smooth movements between these artificial bones.

Humans modulate both the force and stiffness of their joints to engage in physical interactions and adapt to situation-dependent requirements. For this reason, we have incorporated the concept of constrained artificial ball joints to provide distinct properties at each individual joint of the hand. This solution involves the integration of a mechanism that modulates the level of friction between the head and base surfaces (Fig. 2), according to the degree of hand closure. In particular, the MCP joint comprises the base of the proximal phalanx, rigidly connected to the palm surface,

and the head of the metacarpal bone which incorporates a spring. In contrast, the DIP and PIP joints are designed with three components: the base of a distal phalanx, and the head and the shaft of a more proximal phalanx, segmented into two semi-spherical parts connected by a spring.

The proposed mechanism includes two spring heads (Fig. 2) that penetrate into the socket, generating a specific external force F_{ext} according to the spring characteristics. These dislocatable joints permit restricted movement at the frontal and transverse plane, while keeping the full range of motion at the sagittal plane. Adjusting the stiffness of the spring allows for varying force levels, consequently regulating friction among joint components. This solution enables diverse behaviors in the hand with a unified joint mechanism design.

Finally, the human thumb, crucial for gripping and pinching actions, is actuated independently from the other fingers and is mainly commanded by four extrinsic muscles. Notably, the CMC joint of the thumb plays a significant role in facilitating the opposing motion required for various grasps. Recognizing the importance of this more proximal joint, the MCP joint of the artificial thumb was designed as a passive joint, that can be adjusted by a human operator instead of actively moved with the motor. This passive movement ensures independence from the under-actuated actuation architecture, allowing for an increased level of opposition.

B. Physical Implementation

Fig. 3 provides an isometric view of the CAD model of the hand, featuring key components and a detailed illustration of the ball joint implementation. The proposed artificial hand is designed with 15 degrees of freedom (DoFs), from which 14 are actively controlled by a single motor. The remaining degree of freedom corresponds to the passive movement of

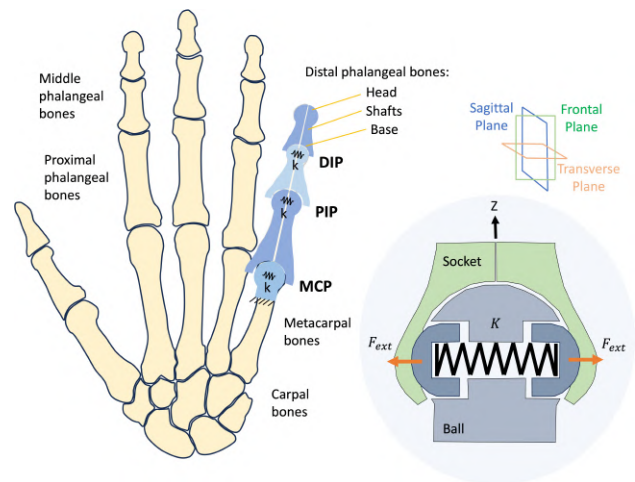


Fig. 2. Bio-inspiration and schematic representation of the proposed concept. Each human joint encompasses a head, a shaft and a base. The right side shows a schematic view of the proposed joint, consisting of three main parts: socket, ball and joint spring. Note that two spring heads are connected to the spring, which creates a pushing force F_{ext} . DIP = distal joint; PIP = proximal joint; MCP = metacarpal joint. k refers to the elastic constant of the joint spring.

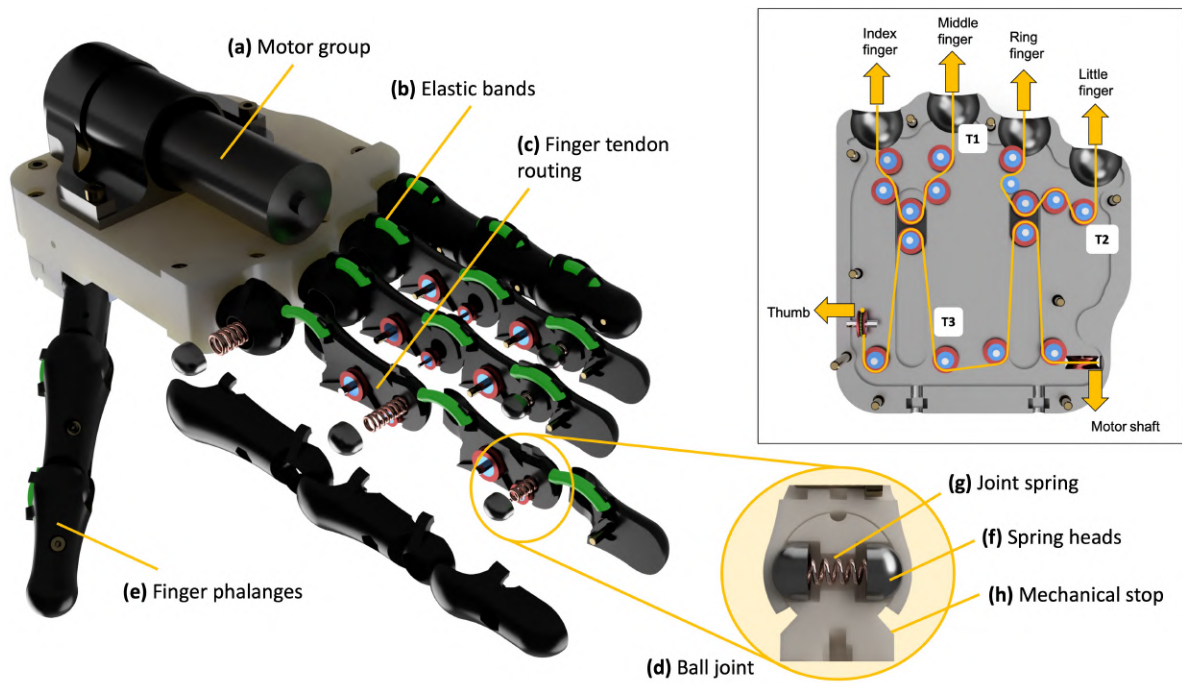


Fig. 3. Proposed system architecture. The hand embeds the (a) motor group within the dorsal part of the palm. A unique finger tendon routing (c) connects all the phalanges of the fingers (15 DoF) to the motor (1 DoA). Constrained artificial ball joints (d) are used to connect the finger phalanges (e) together with several elastic bands (b). The ball joint includes an internal spring (g) to provide specific joint behaviour to each DoF, with two spring heads (f) that apply constant force to the external walls. Two mechanical stops (h) are designed to limit the lateral excursion that these fingers present. In the upper right corner, there is an overview of the actuation architecture. Three interlinked tendon routings (T_i) are connected to a singular actuator (motor shaft) for a synchronized closure of the hand. Note that the ending direction of each tendon side is denoted by yellow arrows.

the proximal joint of the thumb, which allows to achieve different grasping configurations. Exploiting the recent trend of soft robotic technologies and coordinated architectures [3], this design incorporates constrained artificial ball joints (d) within the interphalangeal connections of its digits (e), secured by elastic bands (b). This ball socket joint features an embedded spring that applies pressure to the rotation axis. Therefore, adjusting the stiffness of this internal spring increases contact friction. We implemented a unique tendon routing mechanism for finger flexion (c) that connects all DoFs to a single motor (a), thereby achieving the simultaneous actuation of each joint. Fig. 3 (top right) offers a detailed view of the tendon routing layout in the corresponding CAD model of the palm. Note that three tendons (labeled as $T1$ - $T3$) are linked through two sliders, facilitating synchronized flexion of all fingers and the thumb as the motor rotates. The elastic bands (b) at the dorsal side of the hand permit the finger extension. Specifically, the spherical design of the joints and its friction primarily influences two secondary axes of rotation, when subjected to external forces, leaving the main finger axis responsible for finger flexion actuated by the motor. The constrained joints and their stiffness properties mainly impact the force required to flex the fingers through the tendon pulled, and influence the lateral deformation of the fingers, which significantly affects the final geometry of the hand upon contact. The incorporation of constrained artificial ball joints offers adaptable features while enabling the fine-tuning of movement restriction. The latter is crucial

to attain repetitive trajectories for specific fingers involved in precision gripping, such as the index and thumb. The joint springs (g) were selected to achieve varying levels of pushing force from the spring heads (f) and create distinct levels of friction between the string heads and socket surfaces (see Fig. 3(d)). This feature expands the manipulability ellipsoid without compromising the anthropomorphic form and functionality of the hand.

Embedded intelligent machines exhibit an autonomous character, emphasizing a balance between morphology, materials, and control design [25]. This principle is harnessed in the present design through the use of a differential actuation mechanism, which enables its adaptable finger behavior. The adaptability to the object's shape permits larger contract areas of fingers and the execution of several grasping geometries upon encountering external forces.

A prototype of the proposed hand was built to validate the grasping capabilities (Fig. 1). The hand is actuated with a Maxon DC-X-19S 12 V motor equipped with a magnetic encoder. A single Dyneema tendon is routed through all fingers and joints and wrapped around pulleys with radius of 2 mm. The electronic units hosting motor control, sensor processing, and communication are located in the dorsal part of the hand, while the battery is placed externally. During the grasping experiments, the hand wore a commercial working glove with padded rubber surfaces, to guarantee contact compliance and grip. The palm and phalanges are 3D printed using PLA and ABS, respectively. The overall mass of the

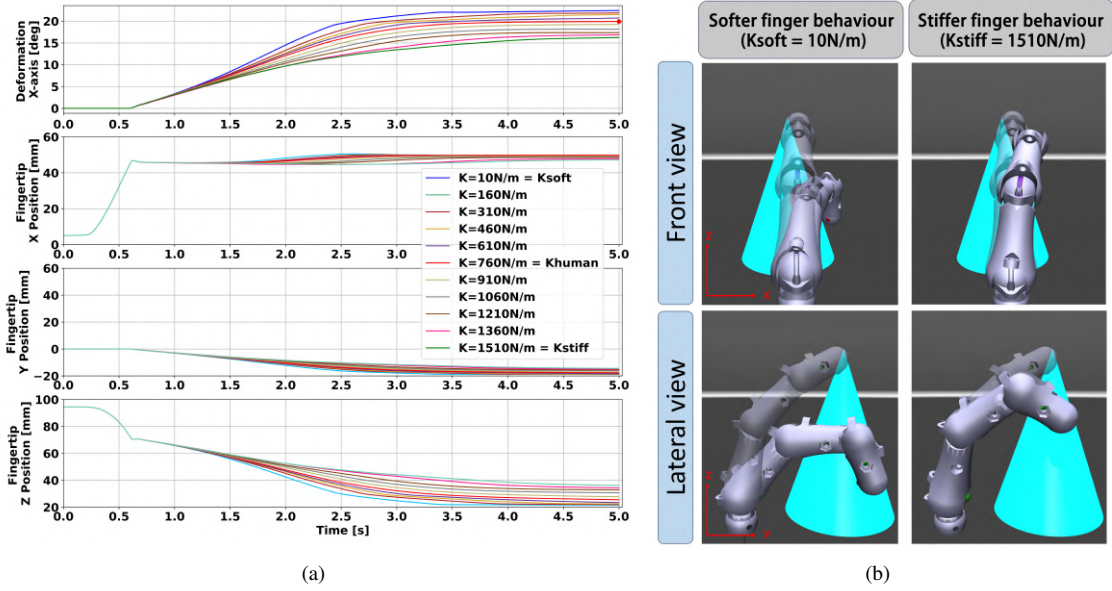


Fig. 4. Simulated workspace analysis. A CAD model of the finger is imported into Mujoco, and diverse stiffness characteristics at the joint are assessed. A cone object is employed to evaluate finger deformation with contact at various positions. Panel (a) shows the temporal evolution of the fingertip position for diverse tested stiffness values. On the top row, the max deformation of the fingertip is reported. First contact with the object occurs at $t=0.612s$. Panel (b) illustrates the comparative behaviours of a finger during simulation with stiffness settings of $K_{soft} = 10N/m$ and $K_{stiff} = 1510N/m$, showcasing how different stiffness levels impact finger flexure.

hand is 363g and the size is 8.65x18.5x2cm.

The outlined architecture ensures a simplified control interface while providing the benefits inherent to a poly-articulated hand. The softness inherent in this articulated robotic device enhances system robustness, enabling extreme finger configurations without the risk of damage, thanks to the dislocatable nature of the joints.

III. EXPERIMENTAL VALIDATION

A. Simulated Workspace Analysis

We explored a set of simulated finger configurations using the CAD model in Mujoco, enabling the assessment of diverse stiffness characteristics at the joints. The simulation involved flexing the finger model while making contact with a cone-shaped object, whose sloping surface provides a continuous point contact and controlled lateral movement as the finger descended along the z-axis. The temporal evolution of fingertip positions within the simulated environment (Fig. 4) guided the determination of desired stiffness levels in the system. The maximum lateral deformation of a human finger (19.9 degrees on the x-axis) [26] served as a reference for achieving a balance between precision and adaptability in the robotic finger design. Through eleven simulations at varying stiffness levels, it was found that at the stiffness of $k = 10N/m$, the finger lateral displacement closely approximated 19.9 degrees at the final phase of the simulation cycle (5 seconds). Thus this was chosen as the reference value for simulating human finger properties. Force at first contact is equal to 0.090N (on x-axis), while force ($t=5s$) is 0.378N. To improve precision and reproducibility in the thumb and index fingers trajectory, a higher stiffness $K_{stiff} = 1510N/m$ was assigned. Fig. 4(b) shows the behaviour of the finger under

this setting, which exhibits minimal displacement along the x-axis upon contacting a sloped surface. A lower stiffness $K_{soft} = 10N/m$ was assigned to the middle, ring and little fingers, to achieve high adaptability while grasping large objects. It is important to emphasize that the stiffness values K_{soft} and K_{stiff} were selected not as specific parameters for actual application but to visualize and demonstrate different responses of fingers in the simulation environment. These findings led the choice of stiffness in the physical joints, together with experimental data and specific manufacturing constraints.

B. Manipulation Taxonomy

Artificial hands exhibit considerable diversity in both form and function, which strongly affect their grasping and interaction capabilities. To quantitatively assess and compare the manipulation capabilities of the proposed terminal device, a taxonomy, inspired by [24], is reported in Fig. 5 and in the video attachment¹. The initial spring characteristics selected for each finger was based on two primary desired conditions: maximizing the range of possible grasp geometries in the taxonomy tree and closely resembling geometries achieved with the human hand. Results demonstrate that all 16 grasp types of the taxonomy are feasible with our device, and that multiple finger parts, beyond just the fingertips, make contact with the object, offering particular advantages for power grasps. Notably, the passivity of the proximal joint of the thumb plays a significant role in controlling the closure orientation of this finger, increasing the level of opposition for securing the object. For this design, the contact

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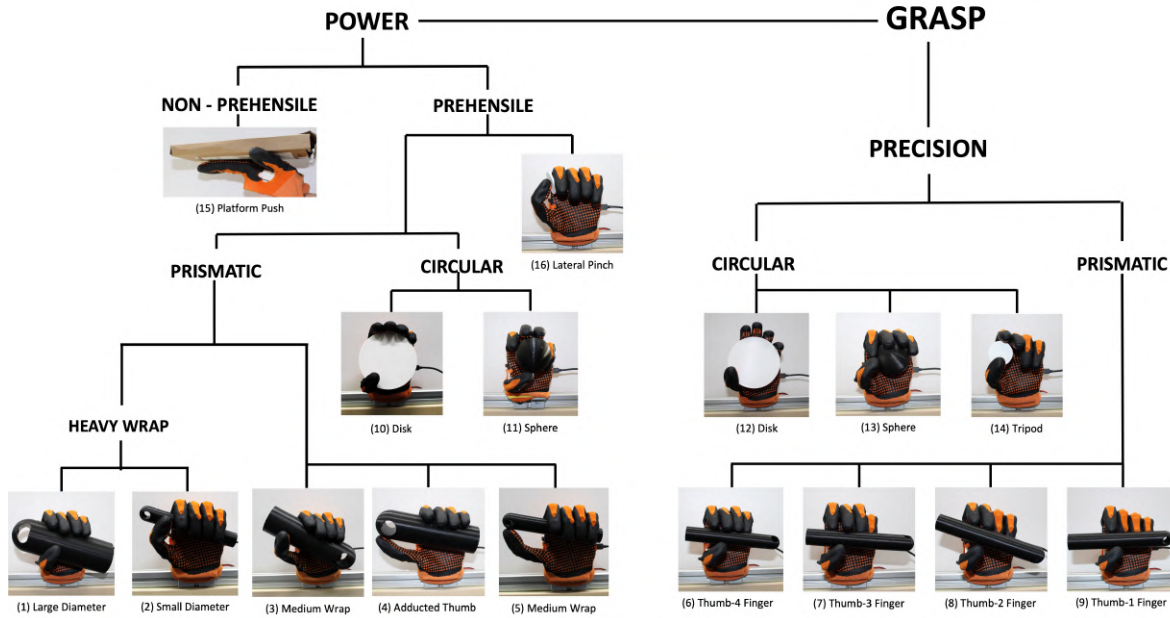


Fig. 5. Manipulation taxonomy, based on [24]. The grasp types are organized along the vertical axis, progressing from detailed to gross tasks and geometry. On the horizontal axis, the arrangement shifts from increased power and object size to increased dexterity and decreasing object size.

with the palm is also of paramount importance due to the adaptability of the fingers upon contact and the strategic use of the palm to prevent the object from moving during its manipulation. Note that the predetermined trajectory for each finger introduces challenges in precisely controlling the movement of specific phalanges during free motion. This is particularly evident in *precision - circular* grasp, where the tip of the thumb may not consistently make contact with the disk or tripod.

C. Functional grasping

Fig. 6 reports a group of functional objects grasped with the proposed system, presented also in the video attachment. The findings underscore the considerable diversity in final hand geometries achieved with the proposed terminal device, which corresponds to the applied contact forces in relation to object stiffnesses and shapes. These range from precise grasping, as demonstrated with a key (Fig. 6(a)), where contact is limited to the fingertips of the thumb and index, to a more tripod-like gripping, observed in the case of the strawberry (Fig. 6(d)) and duck (Fig. 6(g)). A lateral grip is evident when grasping the card (Fig. 6(b)), with the thumb in reposition. Objects from the third column in Fig. 6 exemplify cases where various parts of the fingers and hand make contact with the object, indicative of a power grasp type. Note that, as for humans, the involvement of the little finger contributes to the grasp opposition quality. Furthermore, palm contact is also experienced in cases involving large spherical objects, such as the apple (Fig. 6(c)). Finally, the grape showcases (Fig. 6(i)) a firm grip on a delicate object without exerting undue force.



Fig. 6. Grasp examples of function objects. Common objects used for daily activities were tested, with a subset of nine objects specifically chosen to showcase various aspects of hand functionality. This selection underscores distinct grasps ranging from precise and lateral to power, each of which corresponds to a column. Among them, finger contact adjusts to the forces exerted during the physical interaction with the object.

IV. DISCUSSION & CONCLUSIONS

This work introduced a novel terminal device with constrained ball joints that interconnect multiple finger pha-

langes and a selectable level of stiffness. This design relies on objects to act as constraints, restricting finger flexion to enclose the grasped objects. We preliminarily characterize the joint resistive forces across various axes for different internal spring values in simulation. Results suggest improved repeatability of finger flexion trajectories resulting from the stiffer joint characteristics which report minimal lateral deformation.

Results in [27] suggest that designers consider interaction surfaces of the hand/grip other than fingertips. This emphasizes the importance of hand parts geometry and embedded materials. Consequently, terminal devices should balance both prehensile and non-prehensile capabilities. In this context, the inclusion of constrained ball joints, soft articulated robotic technologies and the developed differential actuation mechanism offer features at opposite ends of this spectrum, providing distinct manipulation benefits. These advantages range from more precise and repeatable grasping to enhanced adaptability of the finger components for securing various objects. The hand adaptability and synchronized finger movement allow to achieve a large number of grasping actions (Fig. 5), but introduces limitations in terms of individual finger movements.

Results illustrated in Fig. 6 reveal a wide range of hand geometries and manipulation capabilities exhibited by the proposed system when interacting with a diverse set of common objects. A salient feature is the variations observed in the level of opposition of the thumb and the contact points of the fingers, whether at the fingertips or more proximal phalanges. Furthermore, another feature is particularly evident in cases such as the duck and apple, where the middle and ring fingers exhibit lateral deformation to better adapt to the object's shape and respond to external forces. This feature differs from the expected behavior of conventional hands, designed with rigid linkages.

Future works will include a physical characterization to complement the insights from simulations and offer more detailed information on the motion principles of the proposed joints. Moreover, the system will be systematically tested to assess usability and robustness, with a specific emphasis on its intuitiveness for operators and applicability in real-world scenarios. Further developments of the proposed design will aim to adjust the joint stiffness using a microcontroller and optimize it for various types of grasps, including the actuation of the thumb opposition, currently passive.

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